REPLACE AND MODERNIZE LEGACY VPN ARCHITECTURES WITH ZSCALER ZERO TRUST

NAAN MUDHALVAN PROJECT

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INTRODUCTION

- **Traditional VPNs provide full network access once authenticated.**
- **Legacy VPN architectures are slow, complex, and insecure.**
- **Modern organizations need secure, cloud-native access.**
- **Zero Trust is the modern approach to prevent security risks.**

PROBLEM STATEMENT

Challenges with legacy VPNs:

- **Over-permissioned network access**
- **Large attack surface**
- **Exposure to internet threats**
- **Poor user experience due to traffic routing**
- **High IT operational overhead**

PROJECT OBJECTIVE

This project aims to:

- **Replace legacy VPNs with Zscaler Zero Trust**
- **Reduce attack surface and prevent lateral movement**
- **Secure application-level access for users**
- **Improve remote user performance**
- **Simplify IT management with cloud-native tools**

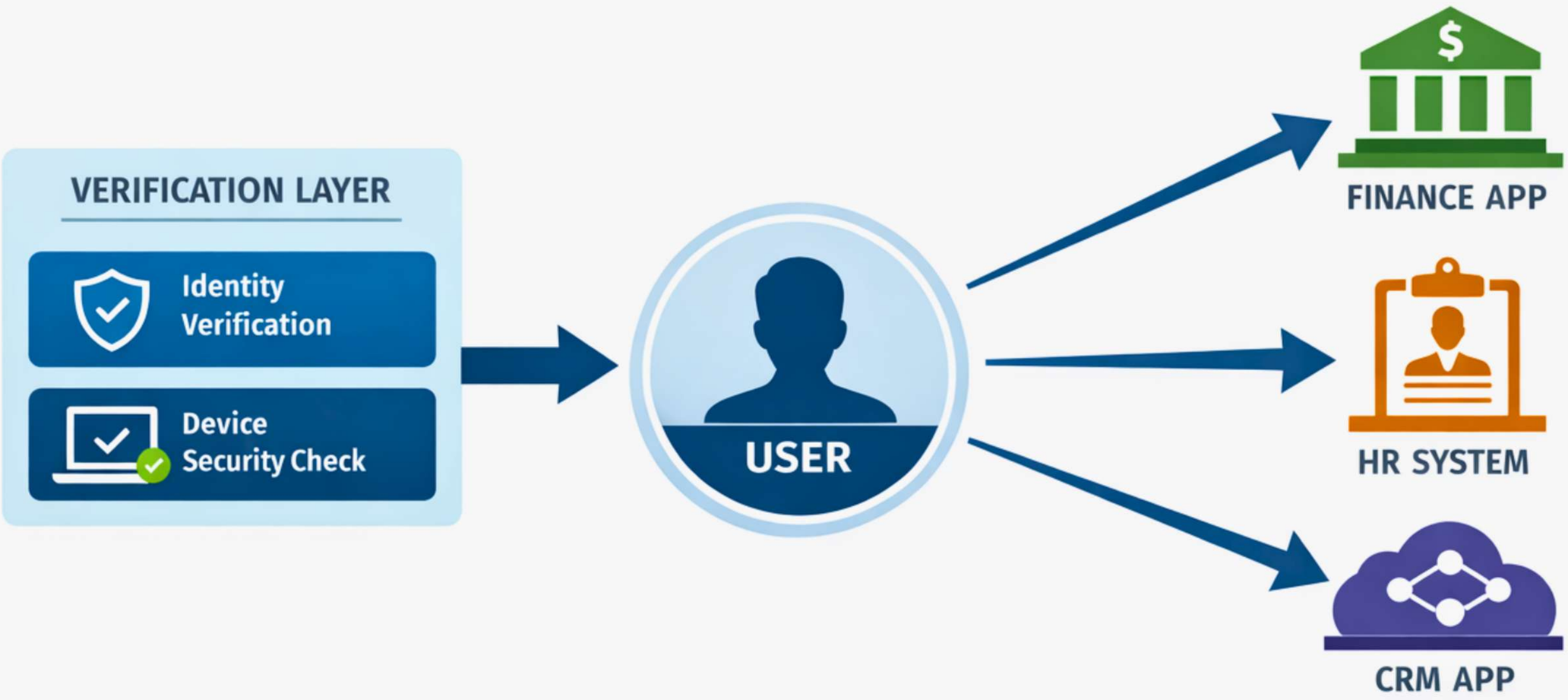
WHAT IS ZERO TRUST

ZERO TRUST PRINCIPLE: “NEVER TRUST, ALWAYS VERIFY”

Key Points:

- **Verify identity before granting access**
- **Apply least privilege policies**
- **Continuous authentication**
- **Secure user-to-application connections**

Zero Trust Principle – Never Trust, Always Verify



Only Verified Users Access Specific Apps

ZSCALER ZERO TRUST SOLUTION

Zscaler provides cloud-native security:

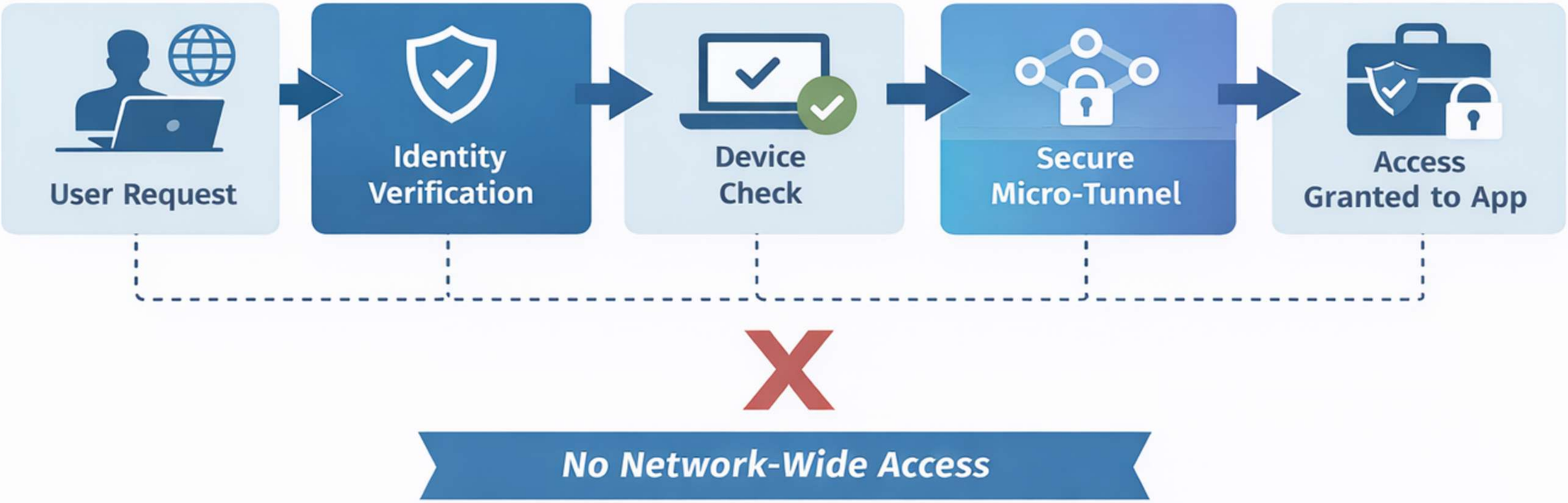
- **Zscaler Private Access (ZPA)**
- **Application-specific secure access**
- **Identity-based authentication**
- **Direct user-to-app connection (no network exposure)**

HOW ZSCALER WORKS

Workflow:

- **User requests access to an application**
- **Identity verification and device posture check**
- **Secure micro-tunnel established**
- **Access granted only to required application**
- **No access to entire corporate network**

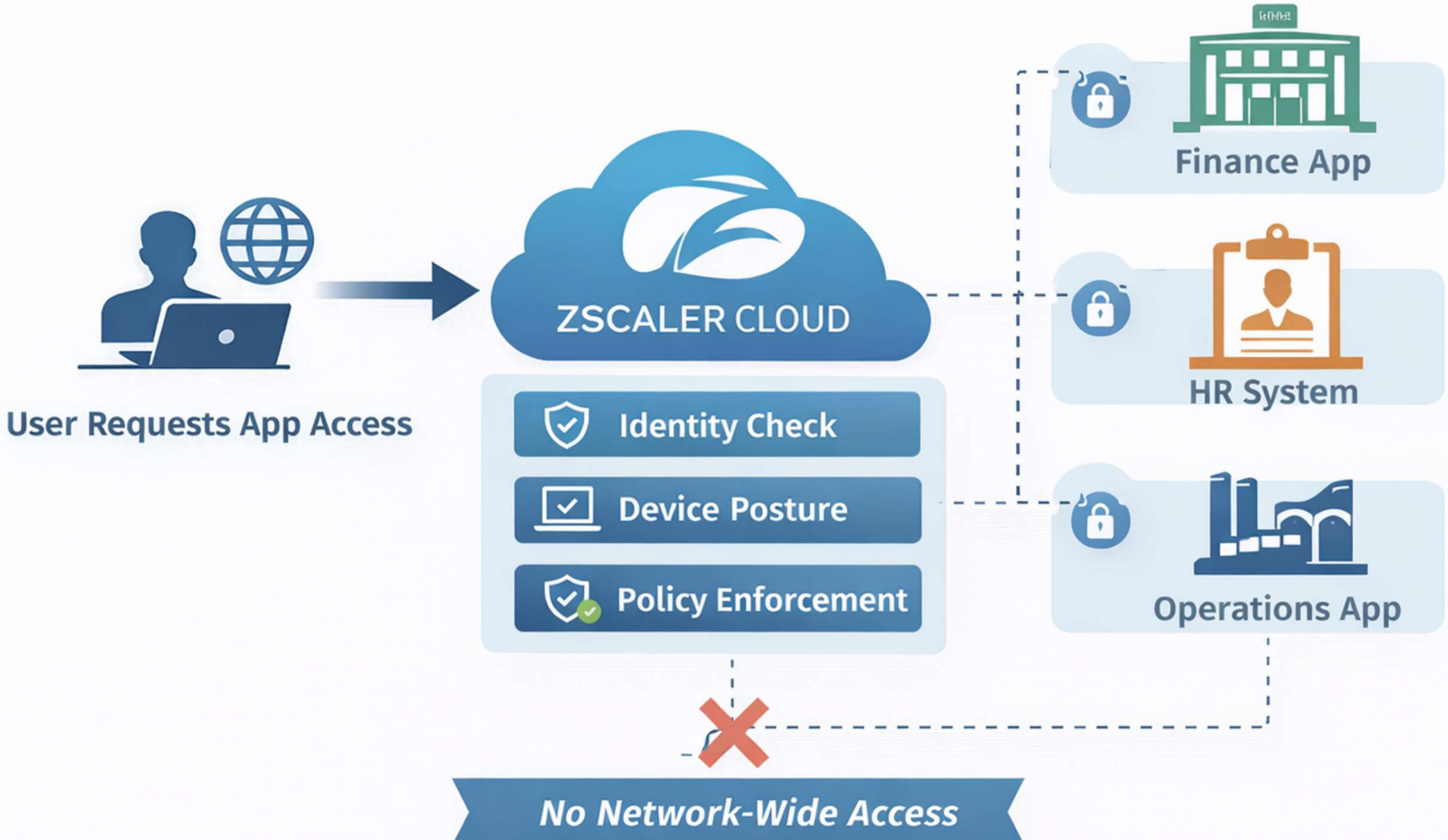
How Zscaler Works: Zero Trust Access Workflow



ADVANTAGES OF ZSCALER ZERO TRUST

- **Minimizes attack surface**
- **Prevents lateral movement**
- **Secure remote and hybrid workforce access**
- **Faster performance for users**
- **Centralized cloud-based management**

Zscaler Cloud-Native Zero Trust Architecture

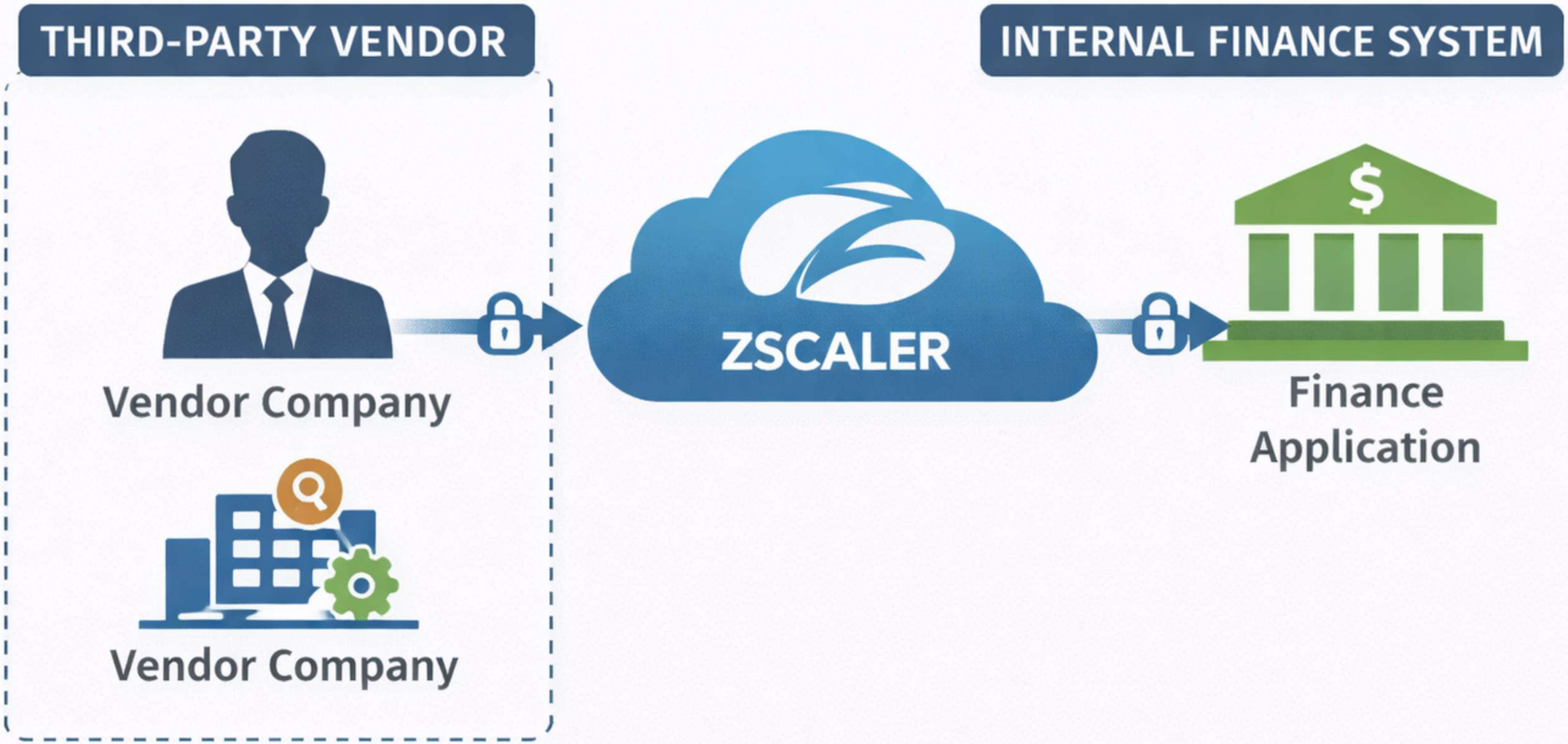


USE CASE EXAMPLE

Scenario: Third-Party Vendor Access

- Vendor requests access to finance system
- Zscaler verifies identity and device compliance
- Access granted only to finance application
- Temporary access policy enforced

Zscaler Zero Trust Access Use Case: Vendor Access



Temporary, Verified Access Only

EXPECTED OUTCOMES

- **Improved organizational cybersecurity**
- **Reduced risk of network-based attacks**
- **Secure hybrid work environment**
- **Simplified IT operations**
- **Future-ready infrastructure**

FUTURE SCOPE

Potential enhancements:

- **AI-powered threat detection**
- **Advanced Data Loss Prevention**
- **IoT and operational technology security**
- **Automated security incident response**

CONCLUSION

- **Legacy VPNs are outdated and vulnerable.**
- **Zero Trust architecture secures application-level access.**
- **Zscaler enables cloud-native secure connectivity.**
- **Project modernizes network security and reduces risk.**